Elfano Sultan Syahbana



Automation & Software Engineer
Industrial IoT · Robotics · Applied AI · Enterprise Software

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SUMMARY

Automation & Software Engineer working across industrial automation, Industrial IoT, robotics, applied AI, and enterprise software. Bridges industrial operations and enterprise software through practical engineering and business process analysis. Supports digital transformation of irrigation operations through Industrial IoT telemetry, has developed autonomous mobile robot and computer vision prototypes at Toyota R&D, and is the sole developer of an enterprise catering platform covering ordering, procurement, inventory, and warehouse operations for a business producing approximately 8,000 meals per day.

Business Process Analysis · Requirements Gathering · Full-Stack Web Development · System & Database Design · Industrial IoT & Telemetry · Remote Monitoring · Sensor Integration & Data Acquisition · ROS, SLAM & Autonomous Mobile Robots · Computer Vision (YOLO, TensorRT, Jetson) · Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Telemetry Engineer

Mar 2025 – Aug 2026

PT Alkatec Mandiri Kencana · Rentang Irrigation Modernization Project · Majalengka, Indonesia

- Modernize irrigation network operations by designing and implementing Industrial IoT telemetry systems using Modbus, MQTT, and GSM communication.
- Support the deployment and operation of 52 telemetry stations and 13 telecontrol stations within the Rentang Irrigation Modernization Project.
- Deliver an end-to-end data acquisition and remote monitoring system by integrating field sensors, data loggers, communication gateways, and telemetry infrastructure.
- Support data-driven water management by monitoring water flow, water levels, climatology, and environmental parameters in real time.
- Maintain reliable real-time monitoring, data accuracy, and data integrity across distributed field devices by working closely with hydrology, automation, and GIS teams.

Independent Software Engineer

Ongoing

Commercial Client Projects · Remote · Alongside full-time roles

Deliver production web applications for paying clients, owning the full lifecycle from requirements gathering to deployment and post-launch improvement.

Enterprise Catering Management Platform — Sole developer of a customer-facing platform and enterprise administration system for an industrial catering business producing ~8,000 meals per day.

- Translated operational workflows (ordering, procurement, inventory, warehouse, approvals), mapped directly with stakeholders, into modules: role-based access, order management, inventory and warehouse management, supplier management, purchase requests and purchase orders, and operational dashboards and reporting.
- Streamlined customer ordering with a menu catalog, online ordering, and WhatsApp ordering; automated business notifications through WhatsApp API (Fonnte) integration.
- Designed the system architecture, database schema, UI/UX, and REST APIs; built with Next.js, React, TypeScript, Tailwind CSS, Next.js API Routes, PostgreSQL, Prisma ORM, and Supabase; deployed on Vercel.
- Ship continuous improvements based on operational feedback from the client.

Corporate Websites — Delivered responsive company profile websites for commercial clients, including contractor firms, from requirements and UI/UX design through deployment and client revisions.

Research & Development Intern · Automation, Robotics & Applied AI — May 2023 – May 2024

PT Toyota Motor Manufacturing Indonesia · Karawang, Indonesia

- Built and developed autonomous mobile robot (AMR) prototypes, using ROS for robot software development and implementing SLAM for mapping and autonomous navigation.
- Developed and deployed YOLO-based computer vision models: prepared and labeled publicly available datasets, and trained object detection models with PyTorch and TensorFlow on Google Colab.
- Optimized inference with NVIDIA TensorRT and deployed models on NVIDIA Jetson Nano and Jetson Orin Nano, achieving approximately 47–60 FPS depending on the hardware platform.
- Integrated computer vision with robotics experiments and participated in research and development of autonomous systems, including automated guided vehicles (AGVs).
- Analyzed work-equipment data to support engineering decisions, and defined and reported engineering KPIs to executive management.

ENGINEERING PROJECTS

SLAM on an Autonomous Mobile Robot using LiDAR — 2024

Implemented Simultaneous Localization and Mapping on an AMR, integrating a LiDAR sensor for mapping and localization.

AMR Prototype on NVIDIA Jetson Nano and Jetson Orin Nano — 2023

Built an autonomous mobile robot prototype on embedded GPU computing platforms.

Earlier Work — 2022

Arduino-based fire-fighting robot; digital line tracer with PID control.

TECHNICAL SKILLS

Programming	TypeScript • JavaScript • Python • C/C++ • HTML • CSS
Frontend	React • Next.js • TypeScript • Tailwind CSS
Backend & Database	PostgreSQL • Prisma ORM • Supabase • Next.js API Routes • REST APIs • WhatsApp API (Fonnte)
Version Control	Git • GitHub
Automation & Robotics	ROS • SLAM • LiDAR Integration • NVIDIA Jetson Nano / Orin Nano • Arduino • PID Control
AI & Computer Vision	PyTorch • TensorFlow • YOLO • NVIDIA Jetson Inference • NVIDIA TensorRT • Computer Vision • Model Training • Google Colab
Industrial IoT	Telemetry • Telecontrol • Modbus • MQTT • GSM • Remote Monitoring • Sensor Integration • Data Loggers • Communication Gateways • Data Acquisition
Engineering	Business Process Analysis • Requirements Gathering • System Architecture • Database Design • UI/UX Design • AI-Assisted Development

EDUCATION

Applied Bachelor's Degree (Sarjana Terapan), Automation Engineering Technology — Aug 2020 – Sep 2024

Institut Teknologi Sepuluh Nopember (ITS), Surabaya · GPA 3.41 / 4.00

- Global Project Based Learning (GPbL) with Shibaura Institute of Technology, Japan (2024)
- Research Student, Electric Drive and Power Electronics System Laboratory (2021 – 2023)
- Finalist, Student Creativity Program (PKM): Community Service (2023) and Creativity Initiative (2021)